

Literature Benchmark Brief: Zhang_2021_ApplEnergy

Title: Spatiotemporal wind field prediction based on physics-informed deep learning and LIDAR measurements

Authors: J. Zhang, X. Zhao (2021)

Venue: Applied Energy | **DOI:** 10.1016/j.apenergy.2021.116641

Cluster: Cluster A: Spatio-Temporal Wind/Wake

Core Theoretical Mechanism & Model Formulation

2D incompressible Navier-Stokes momentum and continuity residual loss regularisation (PINN)

Empirical Scale & Experimental Dataset

High-resolution LiDAR wind field scans across turbine nacelle planes

Relevance to IEEE TSTE & Sustainable Energy Systems

Incorporates fluid mechanics partial differential equations into neural network training loss

Identified Gaps & Operational Vulnerabilities

Requires expensive pulsed LiDAR hardware; unscalable to utility-wide SCADA

Role in Manuscript Defense & Reviewer #2 Proof

Provides theoretical analogy for multi-task physics guidance without expensive LiDAR